

Introduction

What is a Database?

A [database](#) is a collection of related data which represents some aspect of the real world.

DBMS (Database Management System):

DBMS is a collection of inter-related data and set of programs to store & access those data in an easy and effective manner.

What is the need of DBMS?

Database systems are basically developed for large amount of data. When dealing with huge amount of data, there are two things that require optimization: **Storage of data** and **retrieval of data**.

Storage: According to the principles of database systems, the data is stored in such a way that it acquires lot less space as the redundant data (duplicate data) has been removed before storage.

Fast Retrieval of data: Along with storing the data in an optimized and systematic manner, it is also important that we retrieve the data quickly when needed. Database systems ensure that the data is retrieved as quickly as possible.

Purpose of Database Systems

The main purpose of database systems is to manage the data. Consider a university that keeps the data of students, teachers, courses, books etc. To manage this data we need to store this data somewhere where we can add new data, delete unused data, update outdated data, retrieve data, to perform these operations on data we need a Database management system that allows us to store the data in such a way so that all these operations can be performed on the data efficiently.

Characteristics of Database Management System

Here are the characteristics and properties of Database Management System:

- Provides security and removes redundancy
- Self-describing nature of a database system
- Insulation between programs and data abstraction
- Support of multiple views of the data
- Sharing of data and multiuser transaction processing

- Database Management Software allows entities and relations among them to form tables.
- It follows the ACID concept (Atomicity, Consistency, Isolation, and Durability).
- DBMS supports multi-user environment that allows users to access and manipulate data in parallel.

Popular DBMS Software

Here, is the list of some popular DBMS system:

- MySQL
- Microsoft Access
- Oracle
- PostgreSQL
- dBASE
- FoxPro
- SQLite
- IBM DB2
- LibreOffice Base
- MariaDB
- Microsoft SQL Server etc.

Classification of DBMS

The following criteria are normally used to classify DBMSs.

- ❖ **Data model**
- ❖ **Number of users**
- ❖ **Number of sites**
- ❖ **Cost**

➤ **Based on the *Data model* being used the DBMS is classified as follows:**

1. **Hierarchical data model**

- Hierarchical data model represents data as hierarchical tree structure.
- There is no standard language for the hierarchical model; most hierarchical DBMS have record-at-time languages
- Legacy applications still run on database systems based on the **hierarchical** and **network data models**.

2. **Relational data model**

- Stores data in the form of a table.
- A relation database is a data driven not design driven
- Maintaining consistency among all applications is easier
- It supports power query language SQL.

3. **Object-Oriented model**

- It is based on the collection of objects
- This database manages objects for multimedia applications and manages data with complex relationships that are difficult to model and process in a RDBMS.

4. **Object-relational DBMS**

- Relational DBMS have been extending their model to incorporate object database concepts and other capabilities

➤ Based on number of *users*, DBMS can be classified as follows:

- **Single-user systems** support only one user at a time and are mostly used with PCs.
- **Multiuser systems** which include the majority of DBMSs, support concurrent multiple users.

➤ Based on *number of Sites* DBMS can be classified as follows:

1) **Centralized DBMS**

- A DBMS is **centralized** if the data is stored at a single computer site.
- A centralized DBMS can support multiple users, but the DBMS and the database reside totally at a single computer site.

2) **Distributed DBMS**

- A distributed DBMS (DDBMS) can have the actual database and DBMS software distributed over many sites, connected by a computer network.

➤ Based on *the cost* DBMS can be classified as follows:

- The DBMS package cost between \$10000 to \$100000
- Single user work with micro computers cost between \$100 and \$3000
- Other few elaborate packages cost more than \$10000.

Database Applications – DBMS

Applications where we use Database Management Systems are:

- **Telecom:** There is a database to keep track of the information regarding calls made, network usage, customer details etc. Without the database systems it is hard to maintain that huge amount of data that keeps updating every millisecond.
- **Industry:** Where it is a manufacturing unit, warehouse or distribution centre, each one needs a database to keep the records of ins and outs. For example distribution centre should keep a track of the product units that supplied into the centre as well as the products that got delivered out from the distribution centre on each day; this is where DBMS comes into picture.

- **Banking System:** For storing customer info, tracking day to day credit and debit transactions, generating bank statements etc. All this work has been done with the help of Database management systems.
- **Sales:** To store customer information, production information and invoice details.
- **Airlines:** To travel through airlines, we make early reservations, this reservation information along with flight schedule is stored in database.
- **Education sector:** Database systems are frequently used in schools and colleges to store and retrieve the data regarding student details, staff details, course details, exam details, payroll data, attendance details, fees details etc. There is a hell lot amount of inter-related data that needs to be stored and retrieved in an efficient manner.
- **Online shopping:** You must be aware of the online shopping websites such as Amazon, Flipkart etc. These sites store the product information, your addresses and preferences, credit details and provide you the relevant list of products based on your query. All this involves a Database management system.

Advantages of DBMS over file system

Drawbacks of File system

- **Data redundancy:** Data redundancy refers to the duplication of data, let's say we are managing the data of a college where a student is enrolled for two courses, the same student details in such case will be stored twice, which will take more storage than needed. Data redundancy often leads to higher storage costs and poor access time.
- **Data inconsistency:** Data redundancy leads to data inconsistency, let's take the same example that we have taken above, a student is enrolled for two courses and we have student address stored twice, now let's say student requests to change his address, if the address is changed at one place and not on all the records then this can lead to data inconsistency.
- **Data Isolation:** Because data are scattered in various files, and files may be in different formats, writing new application programs to retrieve the appropriate data is difficult.
- **Dependency on application programs:** Changing files would lead to change in application programs.
- **Atomicity issues:** Atomicity of a transaction refers to "All or nothing", which means either all the operations in a transaction executes or none.

For example: Let's say Steve transfers 100\$ to Negan's account. This transaction consists multiple operations such as debit 100\$ from Steve's account, credit 100\$ to Negan's account. Like any other device, a computer system can fail let's say it fails after first operation then in that case Steve's account would have been debited by 100\$ but the amount was not credited to Negan's account, in such case the rollback of operation should occur to maintain the atomicity of transaction. It is **difficult to achieve atomicity in file processing systems**.

- **Data Security:** Data should be secured from unauthorised access, for example a student in a college should not be able to see the payroll details of the teachers, such kind of security constraints are difficult to apply in file processing systems.

Advantage of DBMS over file system

There are several advantages of Database management system over file system. Few of them are as follows:

- **No redundant data:** Redundancy removed by data **normalization**. No data duplication saves storage and improves access time.
- **Data Consistency and Integrity:** As we discussed earlier the root cause of data inconsistency is data redundancy, since data normalization takes care of the data redundancy, data inconsistency also been taken care of as part of it
- **Data Security:** It is easier to apply access constraints in database systems so that only authorized user is able to access the data. Each user has a different set of access thus data is secured from the issues such as identity theft, data leaks and misuse of data.
- **Privacy:** Limited access means privacy of data.
- **Easy access to data** – Database systems manages data in such a way so that the data is easily accessible with fast response times.
- **Easy recovery:** Since database systems keeps the backup of data, it is easier to do a full recovery of data in case of a failure.
- **Flexible:** Database systems are more flexible than file processing systems.

Disadvantages of DBMS:

- DBMS implementation cost is high compared to the file system
- Complexity: Database systems are complex to understand
- Performance: Database systems are generic, making them suitable for various applications. However this feature affect their performance for some applications

DBMS Architecture

- The DBMS design depends upon its architecture. The basic client/server architecture is used to deal with a large number of PCs, web servers, database servers and other components that are connected with networks.
- The client/server architecture consists of many PCs and a workstation which are connected via the network.
- DBMS architecture depends upon how users are connected to the database to get their request done.

Types of DBMS Architecture

There are three types of DBMS architecture:

1. Single tier architecture
2. Two tier architecture
3. Three tier architecture

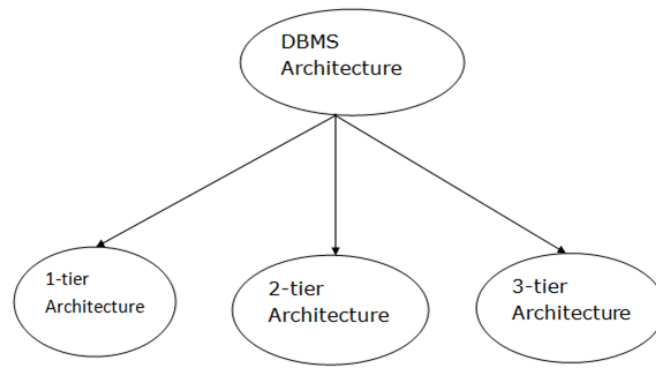


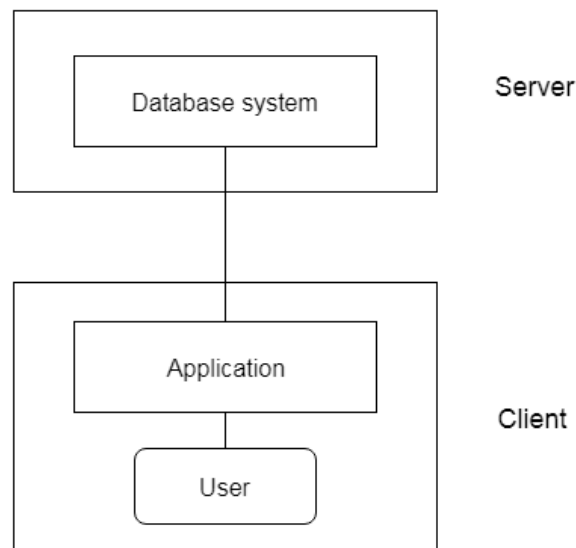
Fig.: Types of DBMS Architectures

1-Tier Architecture

- In this architecture, the database is directly available to the user. It means the user can directly sit on the DBMS and uses it.
- Any changes done here will directly be done on the database itself. It doesn't provide a handy tool for end users.
- The 1-Tier architecture is used for development of the local application, where programmers can directly communicate with the database for the quick response.

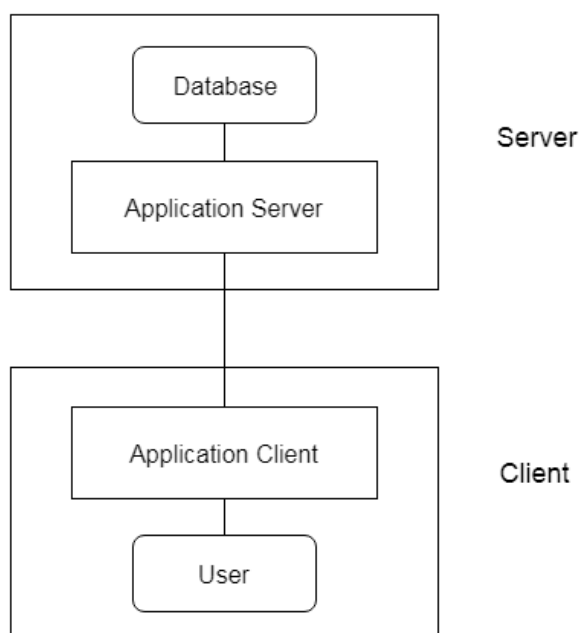
2-Tier Architecture

- The 2-Tier architecture is same as basic client-server. In the two-tier architecture, applications on the client end can directly communicate with the database at the server side. For this interaction, API's like: **ODBC**, **JDBC** are used.
- The user interfaces and application programs are run on the client-side.
- The server side is responsible to provide the functionalities like: query processing and transaction management.
- To communicate with the DBMS, client-side application establishes a connection with the server side.

**Fig: 2-tier Architecture**

3-Tier Architecture

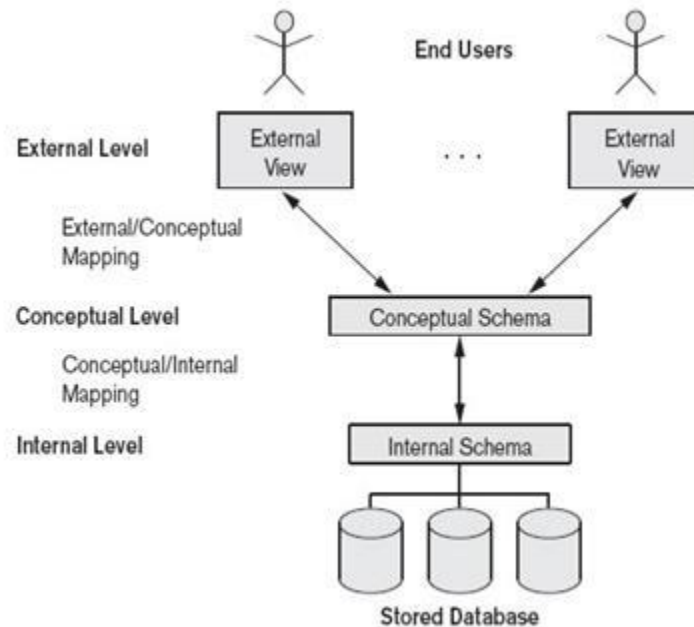
- The 3-Tier architecture contains another layer between the client and server. In this architecture, client can't directly communicate with the server.
- The application on the client-end interacts with an application server which further communicates with the database system.
- End user has no idea about the existence of the database beyond the application server. The database also has no idea about any other user beyond the application.
- The 3-Tier architecture is used in case of large web application.

**Fig: 3-tier Architecture**

View of data in DBMS

The Three-Schema Architecture

The goal of the three-schema architecture, illustrated in Figure is to separate the user applications from the physical database. In this architecture, schemas can be defined at the following three levels:



The three-schema architecture

1. The **internal level** has an **internal schema**, which describes the physical storage structure of the database. The internal schema uses a physical data model and describes the complete details of data storage and access paths for the database.
2. The **conceptual level** has a **conceptual schema**, which describes the structure of the whole database for a community of users. The conceptual schema hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints. A representational data model is used to describe the conceptual schema when a database system is implemented.
3. The **external** or **view level** includes a number of **external schemas** or **user views**. Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group. An external schema is described in a high-level data model.

The stored data *actually* exists is at the physical level only. In a DBMS based on the three-schema architecture, each user group refers to its own external schema. Hence, the DBMS must transform a request specified on an external schema into a request against the conceptual schema, and then into a request on the internal schema for processing over the stored database. If the request is database retrieval, the data extracted from the stored database must be reformatted to match the user's external view. The processes of transforming requests and results between levels are called **mappings**.

Data Independence

Data independence can be defined as the capacity to change the schema at one level of a database system without having to change the schema at the next higher level. We can define two types of data independence:

□ **Logical data independence** is the capacity to change the conceptual schema without having to change external schemas or application programs. We may change the conceptual schema to expand, to change constraints, or to reduce the database.

□ **Physical data independence** is the capacity to change the internal schema without having to change the conceptual schema. Changes to the internal schema may be needed because some physical files were reorganized to improve the performance of retrieval or update.

Database users

End users are the persons who involves in access to the database for querying, updating, and generating reports.

There are several categories of end users:

- Casual end users.
- Naive or parametric end users.
- Sophisticated end users.
- Standalone end users.

■ **Casual end users:** these end users occasionally access the database, but they may need different information each time.

Example: *middle- or high-level managers or other occasional browsers.*

■ **Naive or parametric end users:** these users constantly performs querying and updating the database, using standard types of queries and updates—called canned transactions—that have been carefully programmed and tested.

Example: a) **Bank tellers** check account balances and post withdrawals and deposits. b) **Reservation agents** for airlines, hotels, and car rental companies check availability for a given request and make reservations.

■ **Sophisticated end users:** these end users thoroughly familiarize themselves with the facilities of the DBMS in order to implement their own applications to meet their complex requirements.

Example: *engineers, scientists, and business analysts*

■ **Standalone users:** these end users maintain personal databases by using ready-made program packages.

An example is *the user of a tax package that stores a variety of personal financial data for tax purposes.*

Role and responsibilities of DBA

A database administrator's responsibilities include the following tasks:

- To oversee and manage the resources - In organization where many people use the same resources.
- In a database environment, the primary resource is the database itself. Administering these resources is the responsibility of the database administrator (DBA).

- The DBA is responsible for authorizing access to the database, coordinating and monitoring its use.
- The DBA is accountable for problems such as security breaches and poor system response time.
- Installing and upgrading the database server and application tools.
- Planning for backup and recovery of database information.
- Backing up and restoring databases.
- Generating various reports by querying from database as per need